

RTK base RTCM capture analysis

ArduSimple simpleRTK2B (u-blox ZED-F9P) RTK base · capture 2026-10-08 00:20:24 to 2026-10-08 00:21:23 UTC · 60 one-second epochs · recorded over the board's USB port with PyGPSClient

Summary

- **Clean 1 Hz stream.** Every message type arrives exactly once per second, with every interval 1000 ms and no missed epochs in 60 s.
- **One observation set only.** The USB output carries 1005, 1230 and the four MSM4 observation messages (1074 GPS, 1084 GLONASS, 1094 Galileo, 1124 BeiDou). No MSM7 messages are present.
- **Base is in fixed-position mode.** 31-32 satellites, reported accuracy 8 mm, survey-in not running. Receiver fix type is 5 (time-only), which is the normal report for a fixed-position base. The 1005 station position matches the receiver's own position to 0.0 m.
- **One epoch is about 496 bytes** (478-504), calculated from the message layouts. The same satellites as MSM7 would be about 748 bytes (1.51x).
- **Three of the four observation messages are over 100 bytes** on every epoch, so a radio with a 100-byte packet payload must split them across two RF packets.

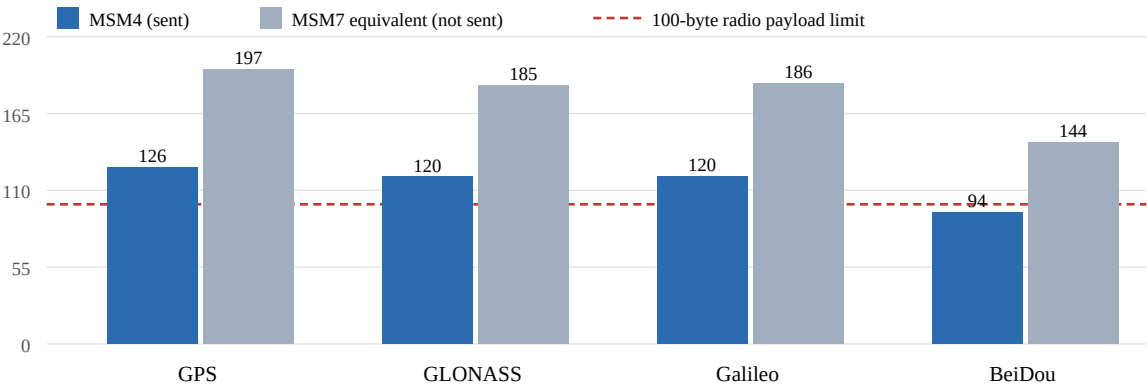
Base receiver state

Item	Value
Position (receiver and 1005 agree)	36.9546203, -121.9746191
Height	-29.29 m ellipsoid / 1.30 m above mean sea level
Satellites used	31-32 (mean 31.8)
Fix type / accuracy	5 (time-only, expected in fixed mode) / 8 mm reported
Survey-in	Not active (fixed-position mode)
Navigation rate / message rate	1 Hz / one of each message per epoch

RTCM messages

Message	Satellites (min-max, mean)	Signal cells (mean)	C/N0 dB-Hz (mean / min)	MSM4 size, bytes (mean, range)	MSM7 equivalent (mean)
1074 GPS	7-9, 8.2	13.0	33.0 / 25	126 (117-134)	197
1084 GLONASS	7-7, 7	12.4	34.4 / 24	120 (111-129)	185
1094 Galileo	7-7, 7	12.6	32.9 / 25	120 (117-123)	186
1124 BeiDou	7-8, 8.0	8.0	36.1 / 26	94 (86-95)	144
1005 station	-	-	-	25 (fixed)	25
1230 GLONASS biases	-	-	-	10 (no biases set)	10

Bytes per observation message (mean per epoch)



Epoch size and time on the wire

Quantity	Value
One epoch, MSM4 plus 1005 and 1230	496 bytes mean (478-504)
Same epoch as MSM7 (calculated)	748 bytes mean (1.51x larger)
Serial time at 115200 baud / 38400 baud	43 ms / 129 ms
Raw RF time at 110 kb/s / 10 kb/s (before protocol overhead)	36 ms / 397 ms
Share of an 11.5 kB/s channel	4.3%

Bytes saved per epoch if one constellation were left out: GPS 126 B (25%); GLONASS 130 B (26%); Galileo 120 B (24%); BeiDou 94 B (19%). GLONASS includes its 1230 message.

Notes

- Message sizes are **calculated** from the RTCM 3.x MSM4 and MSM7 bit layouts using the satellite and signal counts in each message, including the 3-byte frame header and 3-byte CRC. They were not measured on a radio link.
- This is the receiver's output on its **USB** port. The board's correction radio is fed from UART2; the two have been configured to carry the same messages, but this capture does not show what the radio actually delivered.
- Signal levels above are the C/N0 values carried in the observation messages. Satellite counts are per epoch, so the total in use (about 30) is the sum across the four constellations.
- Generated by `analyze_rtcml_log.py` from a PyGPSClient parsed data log.